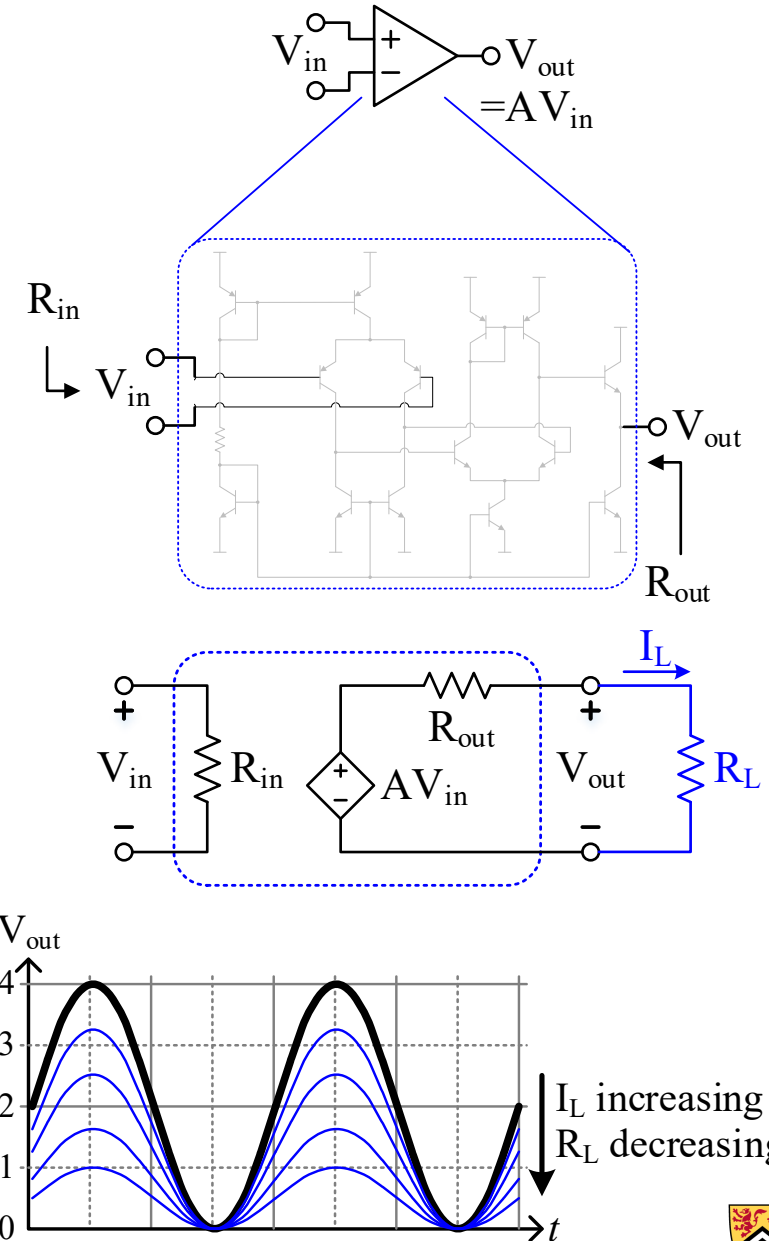


1.2 – Interfaces and Loading

Motivation: Loading

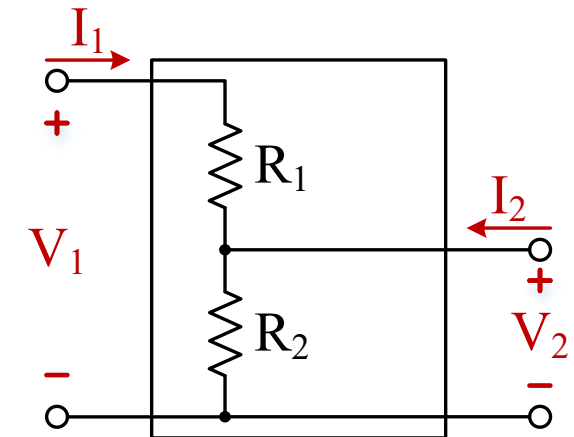
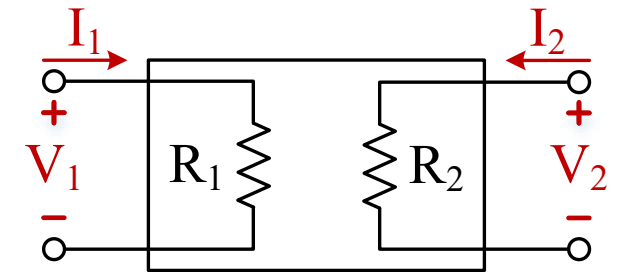
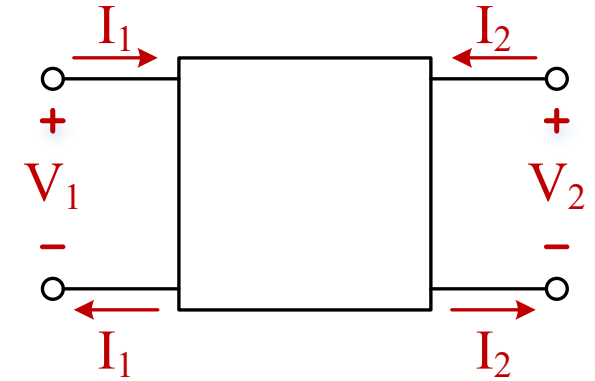
- We need to understand, predict, and mitigate what happens when real $\neq$ ideal
- We need to understand loading, especially when we get to sensors
 - Sensor with high output resistance
 - Amplifier with low input resistance
 - Output transistors can only deliver a certain amount of current
- Result: Output voltage drops as current demands increase
- Model looks like Thevenin-equivalent MTE 120 material
 - Output current losing voltage across output resistance

🔧 Loading Demo



Two-Port Networks

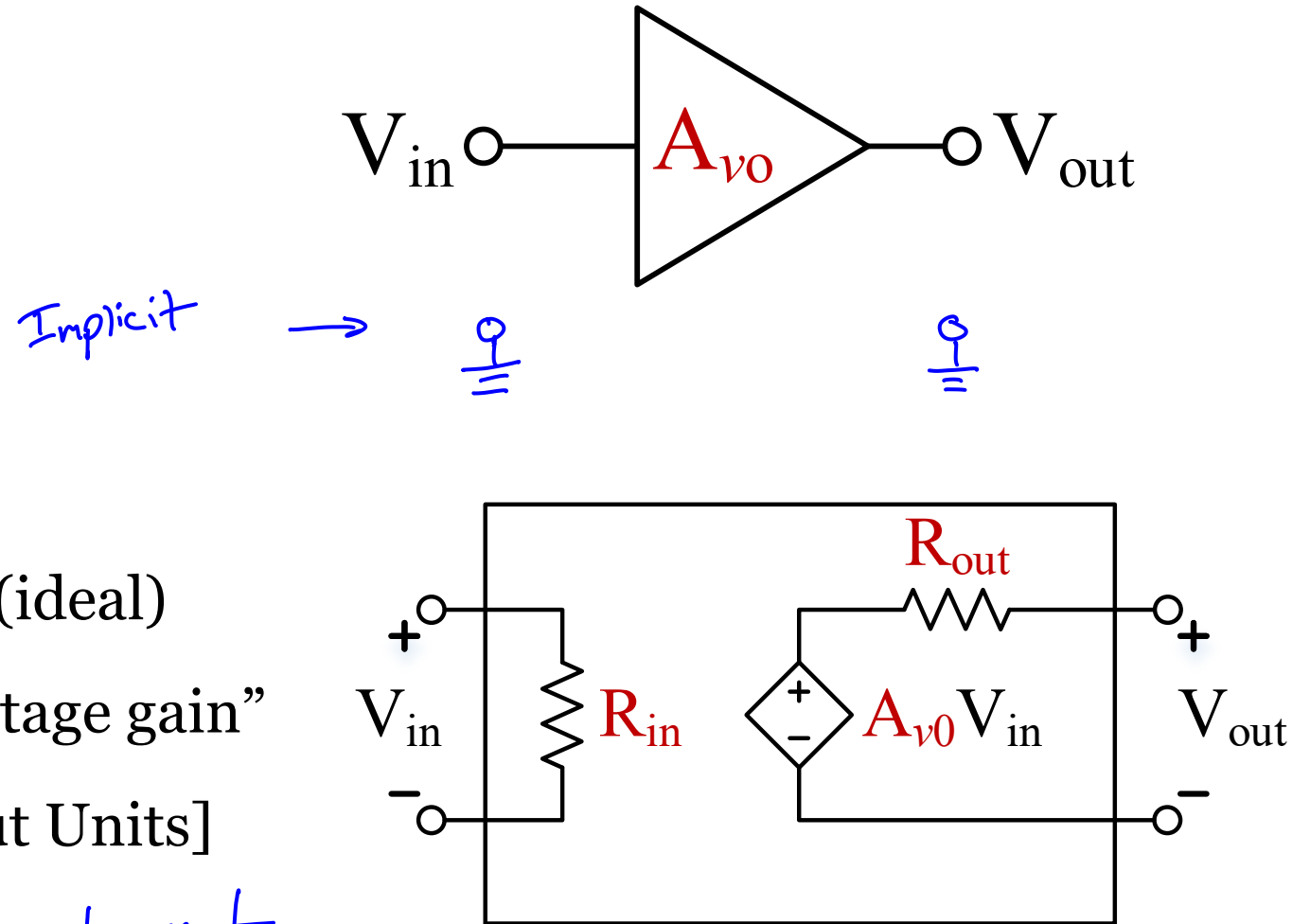
- A circuit with two connections to the outside world
- Sometimes the two ports don't influence each other
- Sometimes they do



Amplifier Modeling

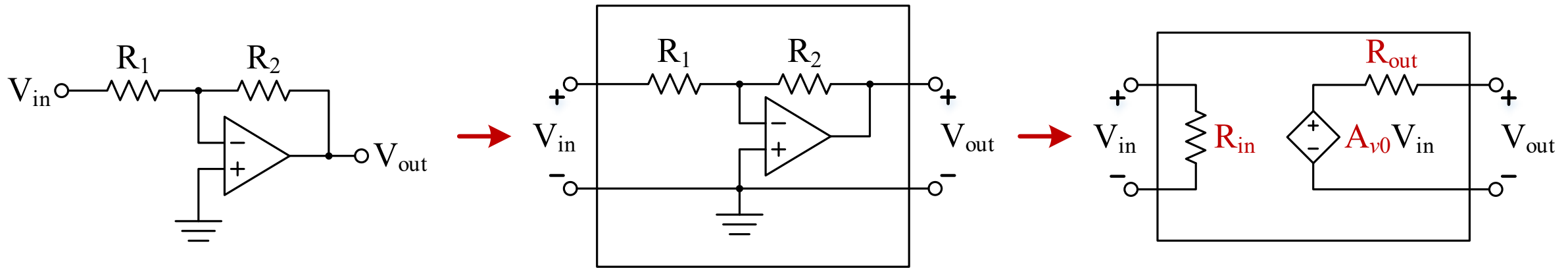
- A two-port network
 - Port 1: Input
 - Port 2: Output
- Input influences output
- Output does not influence input (ideal)
- A_{vo} is called the “open-circuit voltage gain”
 - Units of [Output Units]/[Input Units]

In voltage amplifiers, we do not reduce V/V . It tells you something

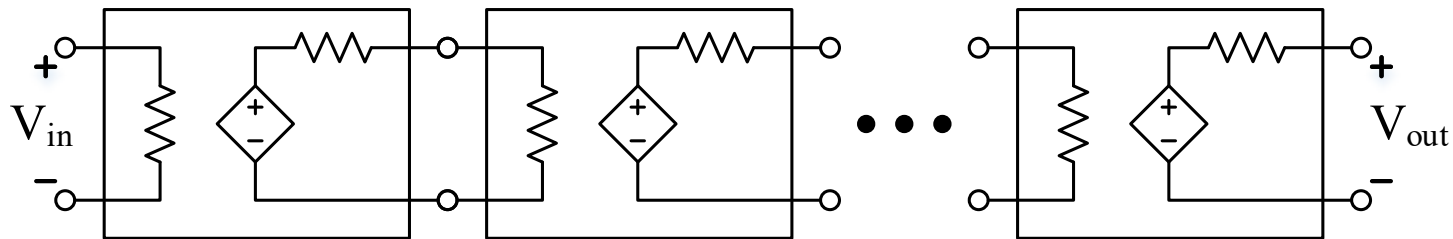


Amplifier Modeling

- Summarizes the terminal behaviour of an amplifier circuit in a simplified model

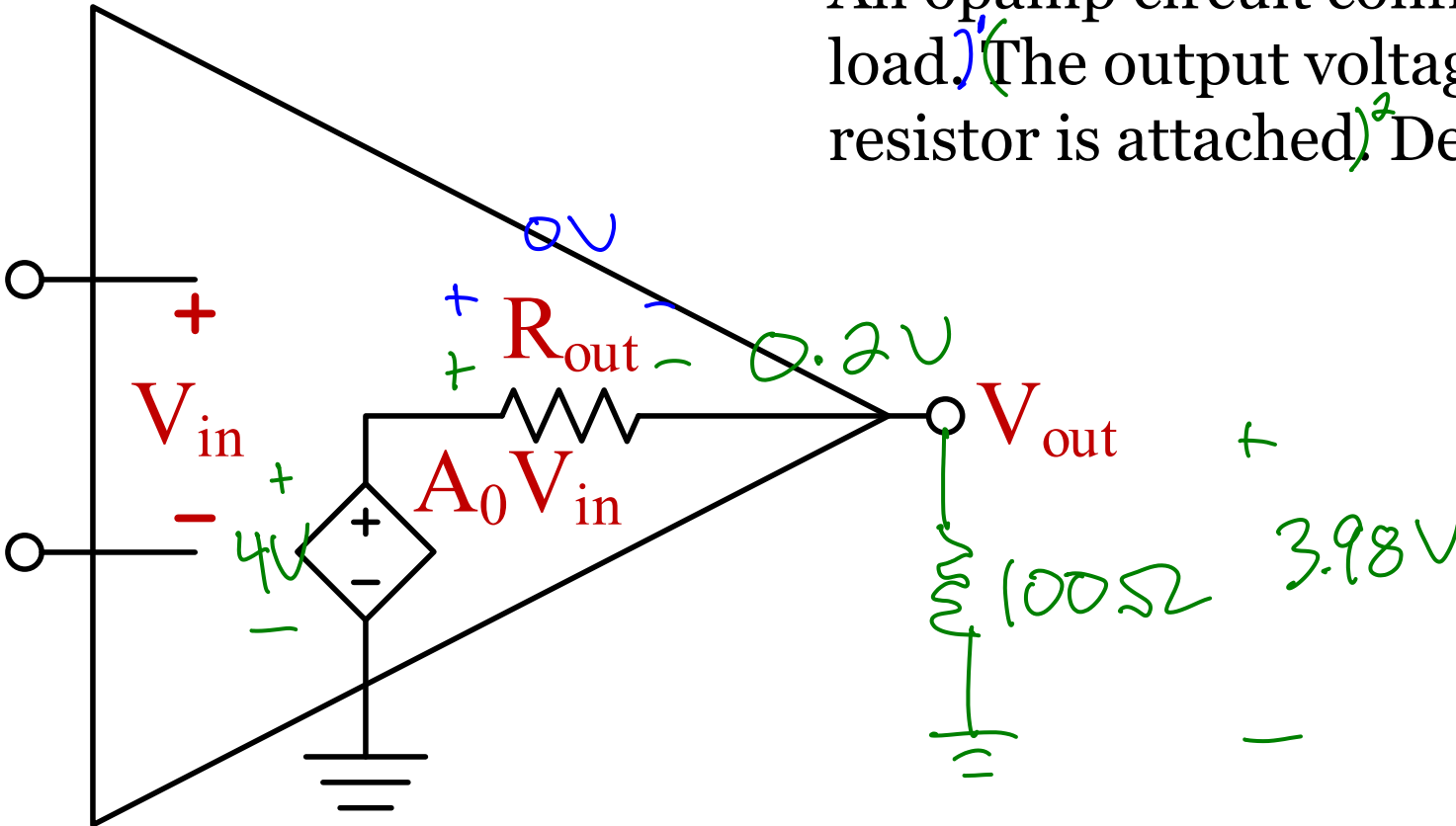


- Makes it straightforward to deal with connecting many amp stages



Example: Output Loading

- An opamp circuit configuration (outputs 4.00 V with no load.) (The output voltage drops to 3.98 V when a 100 Ω resistor is attached).² Determine the amplifier's R_{out} .

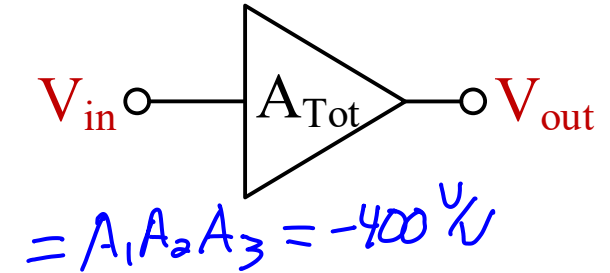
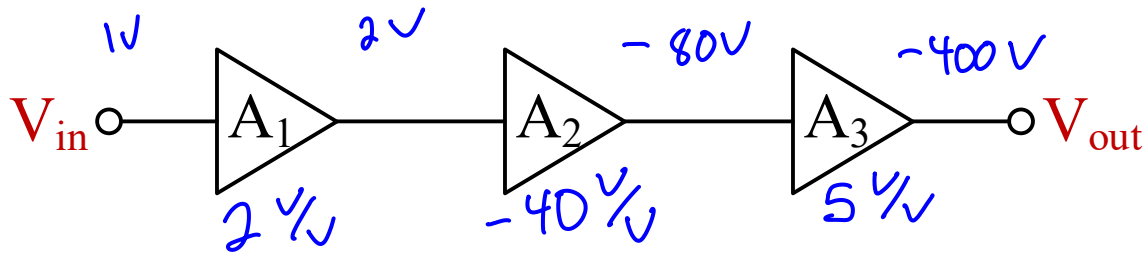


① open-circuit
No current
 $\therefore V_{out} = A_0 V_{in} = 4.00V$
② $V_{out} = 3.98V$
Find R_{out}



Cascaded Amplifiers

- Gains one after another (cascaded) multiply



- Makes it straightforward to deal with connecting many amp stages
- Loading complicates this
- The stages' R_{in} and R_{out} cause signal loss



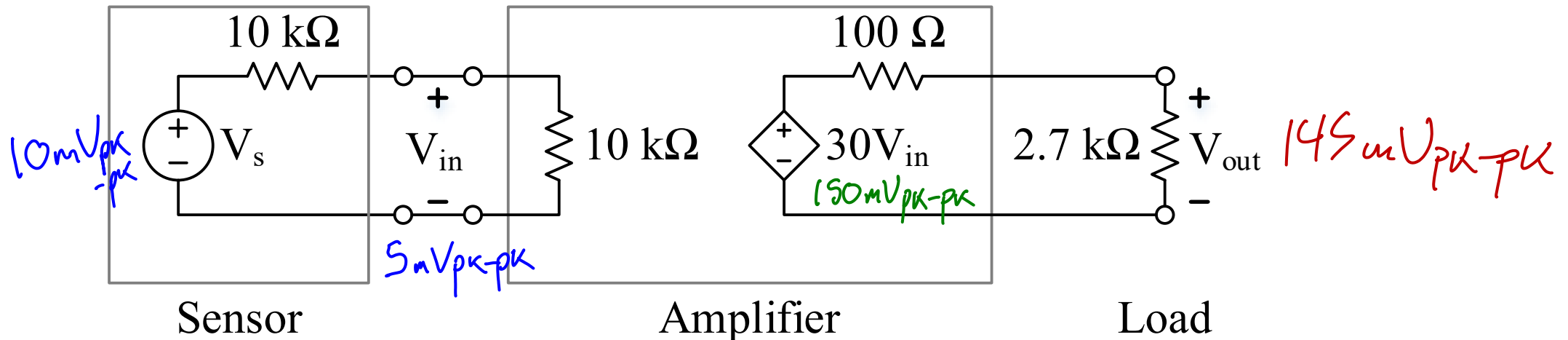
Loaded Amplifier Model Example

- Determine V_{out} when V_s is 10 mV_{pk-pk}

- Where did all the signal go?

$$V_{\text{in}} = V_s \times \frac{10}{10 + 10} = 5 \text{ mV}_{\text{pk-pk}}$$

$$V_{\text{out}} = (30 V_{\text{in}}) \times \frac{2.7 \text{ k}\Omega}{2.7 \text{ k}\Omega + 100} = 145 \text{ mV}_{\text{pk-pk}}$$



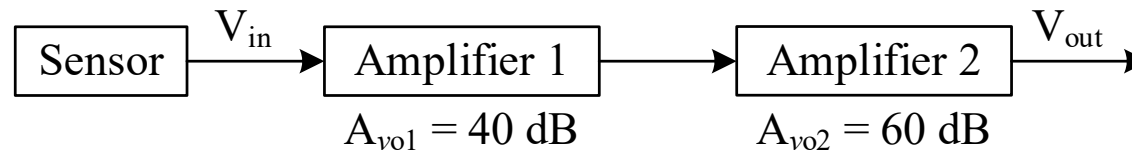
Ratios and Decibels

$$A_{\text{dB}} = 20\log_{10}(|A|)$$

- Great way to express ratios like gains
- Logarithmic instead of linear (add instead of multiply)

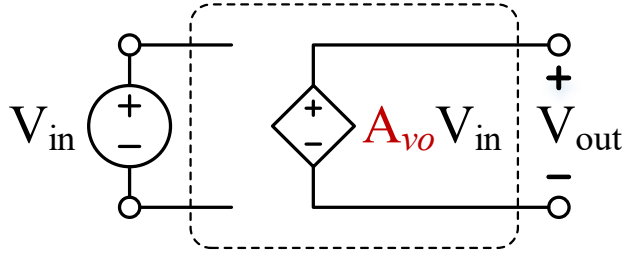
$ A $	A_{dB}	
0.001	-60	Attenuation
0.01	-40	
0.1	-20	
1	0	Amplification
10	20	
100	40	
1000	60	

- **Default assumption: Gains are in [V/V] unless dB is stated**
- Example: Neglecting the effects of loading, determine the open-circuit (unloaded) voltage gain, $A_{vo} = V_{\text{out}}/V_{\text{in}}$ in units of [V/V] and dB ($A_{vo,\text{dB}}$).



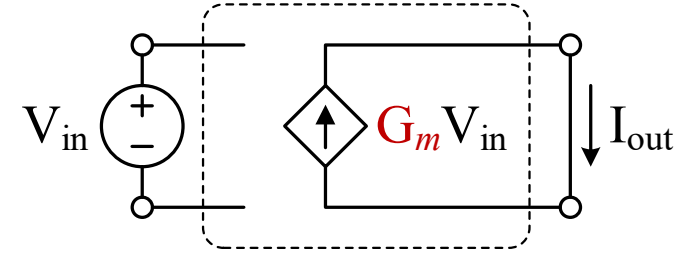
Four Amplifier Types

Voltage Amplifier



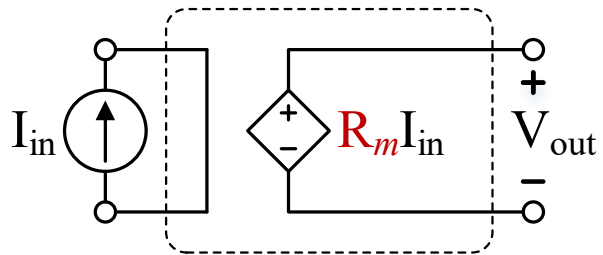
Open-Circuit Voltage Gain: $A_{vo} = \frac{v_{out}}{v_{in}} \left[\frac{V}{V} \right]$

Transconductance Amplifier



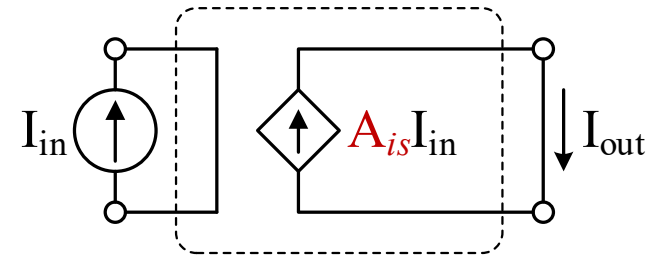
Short-Circuit Transconductance: $G_m = \frac{i_{out}}{v_{in}} \left[\frac{A}{V} \right]$

Transresistance/Transimpedance Amplifier



Open-Circuit Transresistance: $R_m = \frac{v_{out}}{i_{in}} \left[\frac{V}{A} \right]$

Current Amplifier



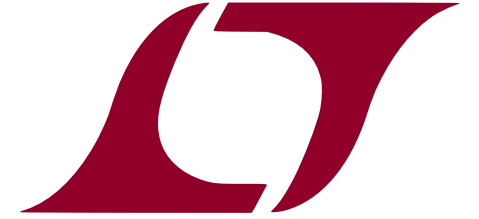
Short-Circuit Current Gain: $A_{is} = \frac{i_{out}}{i_{in}} \left[\frac{A}{A} \right]$



Amplifier Models Summary



LTSpice Modeling Challenge



- Create a model of the source and load in the loading demo
 - The signal went from $840 \text{ mV}_{\text{pk-pk}}$ to $500 \text{ mV}_{\text{pk-pk}}$ when the 4 Ohm speaker was connected
 - Use a switch to show the correct open-circuit and loaded voltage
- Why do you think the switch isn't directly implemented?
 - How else could you handle the switch instead of a voltage-controlled component?
- Can you use the cursors to measure the amplitude with and without the load?
 - Challenge: Can you automate this with .meas statements?
- Express the signal loss in dB

